

Phase offset ambiguities in multi-echo field mapping

Andrew Van

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1 Introduction

Multi-echo field mapping estimates the field from the rate at which phase accumulates across echoes. Every voxel also carries a receive-coil phase offset, present at zero echo time, common to all echoes, and set by coil geometry rather than by the field. This offset varies rapidly in space, steeply near coil elements and discontinuously between receive channels, so measured phase is not spatially smooth even where the field is. Spatial unwrapping assumes that neighbouring voxels differ by less than half a turn, and raw phase violates that assumption. Removing the offset restores the smoothness later processing needs.

The offset is never measured. It must be inferred from the same wrapped phase that it corrupts, and the inference leaves one integer undetermined. This note sets out the consequences of that integer, the rule conventionally used to fix it, the conditions under which the rule fails, and the procedure MEDIC uses to identify and repair those cases.

2 Phase offset estimation

Notation. A quantity is **wrapped** when confined to $(-\pi, \pi]$.

θ, f		coil offset (rad) and field offset (Hz)
φ_e	wrapped	the recorded phase at echo e
$\Delta\varphi$	wrapped	the wrapped phase difference $W(\varphi_1 - \varphi_0)$
$\Delta\varphi_{\text{uw}}$	unwrapped	that difference with its turn count restored
M	integer	the unwrapper's error in whole turns

$\hat{\theta}$	wrapped	the offset estimate
ψ_e	neither	recorded phase less an estimated offset

2.1 Signal model

Acquire E echoes at times $t_0 < t_1 < \dots < t_{E-1}$. Field evolution is linear in time, so the phase at echo e before wrapping is $\theta + 2\pi f t_e$. What is recorded is that value wrapped into one turn. With $W(x) = \arg(e^{ix})$,

$$\varphi_e = W(\theta + 2\pi f t_e). \quad (1)$$

Without W , two echoes would determine θ and f by a straight-line fit. With it the line is known only modulo 2π , and infinitely many (θ, f) pairs remain compatible with the data.

Two properties of W are used below. It is invariant under whole turns, $W(x + 2\pi m) = W(x)$ for integer m , so every real x decomposes uniquely as $x = W(x) + 2\pi m$. Wrapping discards m and unwrapping recovers it. Second, a difference of wrapped values returns the wrapped difference only after a second wrap,

$$W(W(x) - W(y)) = W(x - y), \quad (2)$$

because $W(x) - W(y)$ spans $(-2\pi, 2\pi)$ and coincides with $W(x - y)$ only when it happens to lie within one turn.

2.2 Echo spacing

An echo-planar train produces echoes at constant spacing,

$$t_e = \text{TE}_0 + e \cdot \Delta \text{TE}, \quad e = 0, 1, \dots, E - 1, \quad (3)$$

with TE_0 the first echo time and ΔTE the spacing. Two field scales follow from ΔTE alone. The **wrap spacing** $1/\Delta \text{TE}$ separates neighbouring candidate solutions for the field. The **half-wrap** $1/(2\Delta \text{TE})$ lies midway between them, and serves as the decision boundary for any rule that prefers the smallest field.

2.3 Recovering the offset

The construction below is MCPC-3D-S. Because θ is common to every echo, it cancels in a difference. For the first two echoes,

$$\Delta\varphi = W(\varphi_1 - \varphi_0) = W((\theta + 2\pi f t_1) - (\theta + 2\pi f t_0)) = W(2\pi f \Delta \text{TE}). \quad (4)$$

The middle step follows from Equation 2, since each φ_e differs from $\theta + 2\pi f t_e$ by whole turns. The wrap remains outside the subtraction because $\varphi_1 - \varphi_0$ is a difference of wrapped values.

$\Delta\varphi$ is itself wrapped, but a difference of neighbouring echoes varies slowly in space, so a region-growing algorithm can restore its missing turns. Spatial unwrapping recovers phase differences between voxels rather than absolute turn counts, so what it returns differs from the true unwrapped difference by an unknown whole number of them:

$$\Delta\varphi_{\text{uw}} = 2\pi f \Delta \text{TE} + 2\pi M, \quad (5)$$

with $M = 0$ when the unwrapper is correct. Extrapolating to $t = 0$ then recovers the offset. Writing $k = \text{TE}_0/\Delta \text{TE}$,

$$\hat{\theta} = W(\varphi_0 - k \cdot \Delta\varphi_{\text{uw}}). \quad (6)$$

The outer W is required because $\Delta\varphi_{\text{uw}}$ is unbounded and $k \cdot \Delta\varphi_{\text{uw}}$ with it. M is the only quantity in Equation 6 that the data leaves open.

3 The half-wrap limit

Substituting Equation 5 into Equation 6, and reading the field from the same difference, gives the solution implied by any M :

$$\hat{\theta}_M = W(\theta - 2\pi kM), \quad \hat{f}_M = f + \frac{M}{\Delta \text{ TE}}. \quad (7)$$

The two errors are locked together. The field moves by M whole wrap spacings and the offset by $-2\pi kM$, and neither occurs without the other. The candidate solutions form a discrete family, one per integer, separated by $1/\Delta \text{ TE}$ in field.

Proposition 1. For evenly spaced echoes and any integer M , the pair $(\hat{\theta}_M, \hat{f}_M)$ reproduces Equation 1 identically: $W(\hat{\theta}_M + 2\pi\hat{f}_M t_e) = \varphi_e$ at every echo.

Proof. Working modulo 2π , so that W of either side agrees,

$$\begin{aligned} \hat{\theta}_M + 2\pi\hat{f}_M t_e &\equiv \theta - 2\pi kM + 2\pi(f + M/\Delta \text{ TE})t_e \\ &= \theta + 2\pi f t_e + 2\pi M(t_e/\Delta \text{ TE} - k). \end{aligned} \quad (8)$$

By Equation 3, $t_e/\Delta \text{ TE} = k + e$, so the parenthesis is the integer e , the final term is a whole number of turns, and both sides wrap alike. ■

The cancellation depends on the even spacing. Written as $t_e/\Delta \text{ TE} = k + e$, the ratio k locates the echo train on the $\Delta \text{ TE}$ lattice, and the $-2\pi kM$ carried by the offset is what cancels it. Since the cancellation holds at every echo simultaneously, a longer echo train places no further constraint on M . Unequal spacing would leave $t_e/\Delta \text{ TE} - k$ non-integral at some echo and so produce a detectable residue, but an EPI train is evenly spaced by construction.

3.1 The rule ROMEO applies

No measurement selects M , so the choice must come from elsewhere. ROMEO's global correction, which MEDIC calls, subtracts a single multiple of 2π from the unwrapped phase so that the median rounded turn count over the mask becomes zero. The effect is to select the candidate whose field is smallest,

$$\hat{M} = \operatorname{argmin}_M |f + M/\Delta \text{ TE}|, \quad (9)$$

which is equivalent to requiring $|\hat{f}| < 1/(2\Delta \text{ TE})$.

The rule encodes a physical expectation rather than a measurement. Before imaging, the scanner sets its demodulation frequency to the centre of the water resonance, so the residual field is a small offset rather than an arbitrary one. Where that expectation holds, the rule returns the correct solution. Where it does not, the rule returns an alias, and by Prop. 1 nothing in the data distinguishes the two.

4 Why the rule fails on some frames

A wrong value of M , chosen once and applied to a whole acquisition, would matter very little. Every frame would be displaced by the same $1/\Delta$ TE. The field offset is already referenced to an arbitrary demodulation frequency, so a fixed additive constant changes neither temporal contrast nor spatial structure, and by Prop. 1 it could not be detected in any case.

The difficulty is that M is not chosen once. Three properties of the procedure combine to produce a failure.

The first is that the choice is repeated on every frame. A functional acquisition is processed frame by frame, and each frame resolves M from its own data, without reference to the frames around it.

The second is that the quantity compared against the half-wrap is an estimate. A median rounded turn count is a location statistic over a mask of voxels, each carrying its own field, and it absorbs whatever varies between frames, including physiological change, subject motion, and noise in the voxels the mask admits.

The third is that the comparison has a hard boundary. Far from the half-wrap the margin is wide, and no plausible perturbation of the estimate alters the outcome. Close to the half-wrap the margin approaches zero.

For a subject whose field lies near the half-wrap, M is therefore decided on each frame independently, by a noisy statistic sitting close to a threshold, and different frames fall on different sides of it. Because M is an integer, a frame that falls on the wrong side is displaced by a full wrap rather than by a small amount. The resulting time series is stable to a fraction of a hertz across most frames and displaced by $1/\Delta$ TE on a scattered subset, with no intermediate values.

The damage is temporal. A uniformly wrong field is harmless, but a field that takes two values within a single acquisition corrupts the temporal structure the acquisition was performed to measure.

5 Detecting a failed choice

Prop. 1 appears to rule out any correction. It applies, however, to a **matched** pair, meaning an offset $\hat{\theta}_M$ together with the field $\hat{f}_M$ that accompanies it. The pipeline does not produce the pair in a single step. The offset is formed first, from $\Delta\varphi_{\text{uw}}$ by Equation 6. The field is derived afterwards, by removing that offset and unwrapping in time, and the second unwrapping applies a global correction of its own without reference to the M the offset assumed.

When the two stages disagree, the result falls outside the family in Equation 7. The offset belongs to one candidate and the field to another, so Prop. 1 does not apply to it. This is the situation on a frame where the rule has tipped, because the two stages use different masks and different statistics and can land on opposite sides of the boundary.

Removing the offset from the recorded phase leaves

$$\psi_e = \varphi_e - \hat{\theta}_M, \quad (10)$$

a difference of two wrapped values, and therefore not itself a phase. Unwrapping it in time and fitting

$$\psi_e \approx c + s \cdot t_e \quad (11)$$

gives a slope s , which is the field, and an intercept c , the value the line extrapolates to at $t = 0$. A self-consistent reconstruction has had its offset removed, so its line passes through the origin and $c = 0$.

Proposition 2. If the fit Equation 11 is taken on the solution whose slope is the true field f , then

$$\psi_e \equiv 2\pi f t_e + c \bmod 2\pi, \quad c = W(2\pi k M), \quad (12)$$

with c identical at every echo.

Proof. $\psi_e \equiv (\theta + 2\pi f t_e) - (\theta - 2\pi k M) = 2\pi f t_e + 2\pi k M$, and $2\pi k M \equiv c \bmod 2\pi$. The constant does not depend on e . ■

Two properties make c usable. Being identical at every echo, it displaces the whole fitted line rather than averaging away across echoes. Being a single global constant rather than a per-voxel effect, it can be aggregated over a mask without dilution.

The intercept measures neither the error in the offset nor the error in the field. It measures whether the two are mutually consistent, which is the property the data can check. Its limitation follows from Prop. 1: a matched pair gives $c = 0$, so the intercept cannot rank it against the correct solution. Where more than one candidate remains consistent, the data has been exhausted.

6 The selection algorithm

The procedure runs once per frame. Its output is an integer correction N applied to $\Delta\varphi_{\text{uw}}$ before the offset of Equation 6 is formed, leaving a residual error of $M + N$, and the aim is $N = -M$.

Step 1. Consider $N \in \{-1, 0, +1\}$. A single step is a full wrap of global field, and the global correction has already brought the field inside the half-wrap, so a correction of more than one wrap is not reachable.

Step 2. For each N , shift $\Delta\varphi_{\text{uw}}$ by $2\pi N$, form the implied offset by Equation 6, remove it from both echoes, unwrap in time, fit Equation 11, and record $|c|$ aggregated over a conservative brain mask.

Step 3. Discard any candidate whose $|c|$ exceeds a cutoff. By Prop. 2, neighbouring candidates are separated in intercept by $|W(2\pi k)|$, so the cutoff is placed at half that separation, which makes the test a nearest-neighbour assignment between two possibilities rather than a tuned threshold. The implementation also measures the largest intercept observed among the candidates and adopts the stricter of the two scales, which keeps the test honest when the second unwrapping has already absorbed part of the residue.

Step 4. If one candidate survives, take it, whatever its field. If none survives, change nothing, since a failed fit is not evidence for any particular candidate. If several survive, they are indistinguishable to the data, and Equation 9 selects the one with the smallest estimated $|\hat{f}|$.

The two stages answer different questions, and neither suffices alone. Step 3 uses evidence and rejects candidates the data contradicts. Step 4 uses an assumption, and applies it only among candidates the data has already accepted, which prevents the assumption from overriding the evidence.

Step 4 applies the same rule as Equation 9, so it is worth stating what it changes. The global correction takes an unweighted median of **rounded** turn counts over a dilated mask. Rounding discards the sub-wrap information that indicates how close the decision is, and dilation admits edge voxels whose fields are extreme. Step 4 estimates the field itself, weighted, over a conservative mask. The rule is unchanged; the statistic it is applied to is not.

One degenerate case deserves mention. If k is a whole number then $W(2\pi kM) = 0$ for every M , so the intercept carries no information. Equation 7 then gives $\hat{\theta}_M = W(\theta)$ regardless of M , so no choice alters the output. The implementation detects the zero separation and makes no change, which is the correct behaviour.

7 Estimating the field

Step 4 compares $|\hat{f}|$ against the half-wrap, so the reliability of the comparison depends on what that estimate measures and on how steady it is between frames.

7.1 What the rule refers to

Equation 9 is a statement about **bulk tissue**, since bulk tissue is what the scanner’s frequency adjustment centres. It is not a statement about the mean field, nor about the median field over whatever mask happens to be available.

The field distribution across a brain is asymmetric. It has a narrow peak at the bulk-tissue value together with long tails from air-tissue interfaces, sinuses, signal dropout and mask edges, where susceptibility gradients are large. Any quantile of the whole distribution is displaced toward whichever tail is heavier. Writing the distribution as a mixture,

$$p(f) = (1 - \varepsilon) \cdot p_{\text{bulk}}(f) + \varepsilon \cdot p_{\text{tail}}(f), \quad (13)$$

the rule refers to the centre of p_{bulk} , and the estimator should recover it with ε as small as possible. Two consequences follow.

Enlarging the mask makes the estimate worse. Dilating a brain mask admits precisely the voxels that populate p_{tail} , which raises ε and displaces any quantile of p further from the bulk centre. This reverses the usual expectation that more data improves an estimate.

A bias of a fraction of a wrap is enough to change the decision. The comparison is against the half-wrap, so estimator bias adds directly to the apparent distance from the boundary. A subject whose bulk field already lies near the half-wrap can be carried across it by bias alone, at which point the decision reflects the statistic rather than the data.

7.2 The estimator

MEDIC uses a magnitude-weighted median over an eroded brain mask. Erosion lowers ε geometrically by removing edge voxels, and the weighting lowers it by suppressing low-signal voxels, which are over-represented in the tails. The weight is the magnitude of the second echo, because the field is read from a difference whose noise is dominated by the weaker of the two.

A median is used rather than a mean or an M-estimator. The bulk distribution is broad and right-skewed, and the half-wrap boundary falls within its body rather than out in a tail, so a statistic pulled toward the mean reads systematically closer to the boundary without describing bulk tissue any more accurately.

Continuity is the binding requirement. The estimate decides a threshold comparison that has to give the same answer on consecutive frames of one subject, and an unstable estimator would reintroduce

at this stage exactly the inconsistency described in section 4. A median moves smoothly as the samples move. Estimators defined by an $\arg \max$, including modes in any of their usual forms, do not, since field distributions are frequently multimodal and a near-tie between peaks makes the estimate jump between frames. Stability across frames matters more here than freedom from bias.